1.1 Internet of Things

The Internet of Things (IoT) refers to the billions of physical devices around the world that are now connected to the internet, all collecting and sharing data. Before the advent of IoT, only the computers and mobile phones were capable of connecting to the internet. IoT enables connection of "things" with internet. "Things" can be sensors, actuators and devices containing a microcontroller. These different objects and sensors add a level of digital intelligence to devices that would be otherwise dumb, enabling them to communicate real-time data. The Internet of Things is making the fabric of the world around us smarter and more responsive, merging the digital and physical universes.

The term IoT is mainly used for devices that wouldn't usually be expected to have an internet connection, and that can communicate with the network independently of human action. For this reason, a PC isn't generally considered an IoT device and neither is a smartphone. However, a smartwatch or a fitness band or other wearable device might be counted as an IoT device.

> Do you know?
>
> Initial concept of IoT was to create a separate network of "things". It was not necessary to connect things with the Internet.

1.1.1 Components of IoT

A typical IoT system consists of following main components:

Sensors/Devices

Sensors or devices help in collecting very minute data from the surrounding environment. All of this collected data can have various degrees of complexities ranging from a simple temperature monitoring sensor or a complex full video feed.A device can have multiple sensors. For example, our phone is a device that has multiple sensors such as GPS, accelerometer, camera but our phone does not simply sense things.

Connectivity

The sensors can be connected to the cloud through various mediums of communication such as cellular networks, satellite networks, Wi-Fi, Bluetooth, wide-area networks (WAN), low power wide area network and many more.Each option has some specifications and trade-offs between power consumption, range, and bandwidth. So, choosing the best connectivity option in the IOT system is important.

Data Processing

Once the data is collected and it gets to the cloud, the software performs processing on the acquired data.This can range from something very simple, such as checking that the temperature reading on devices such as AC or heaters is within an acceptable range. It can sometimes also be very complex, such as identifying objects (such as intruders in your house) using computer vision on video.There might be a situation when a user interaction is required.

User Interface

In an IoT systems, the information is made available to the end-user in some way. This can be achieved by triggering alarms on their phones or notifying through texts or emails.A usermight also have an interface through which they can actively check in on their IoT system. For example, a user has a camera installed in his house, he might want to check the video recordings and all the feeds through a web server.However, it’s not always this easy and a one-way street. Depending on the IoT application and complexity of the system, the user may also be able to perform an action that may backfire and affect the system.For example, if a user detects some changes in the refrigerator, the user can remotely adjust the temperature via their phone.There are also cases where some actions take place automatically. By establishing and implementing some predefined rules, the entire IoT system can adjust the settings automatically and no human has to be physically present.In case any intruders are sensed, the system can generate an alert not only to the owner of the house but to the concerned authorities. Figure 1.1 shows user dashboard of an IoT enabled smart home.

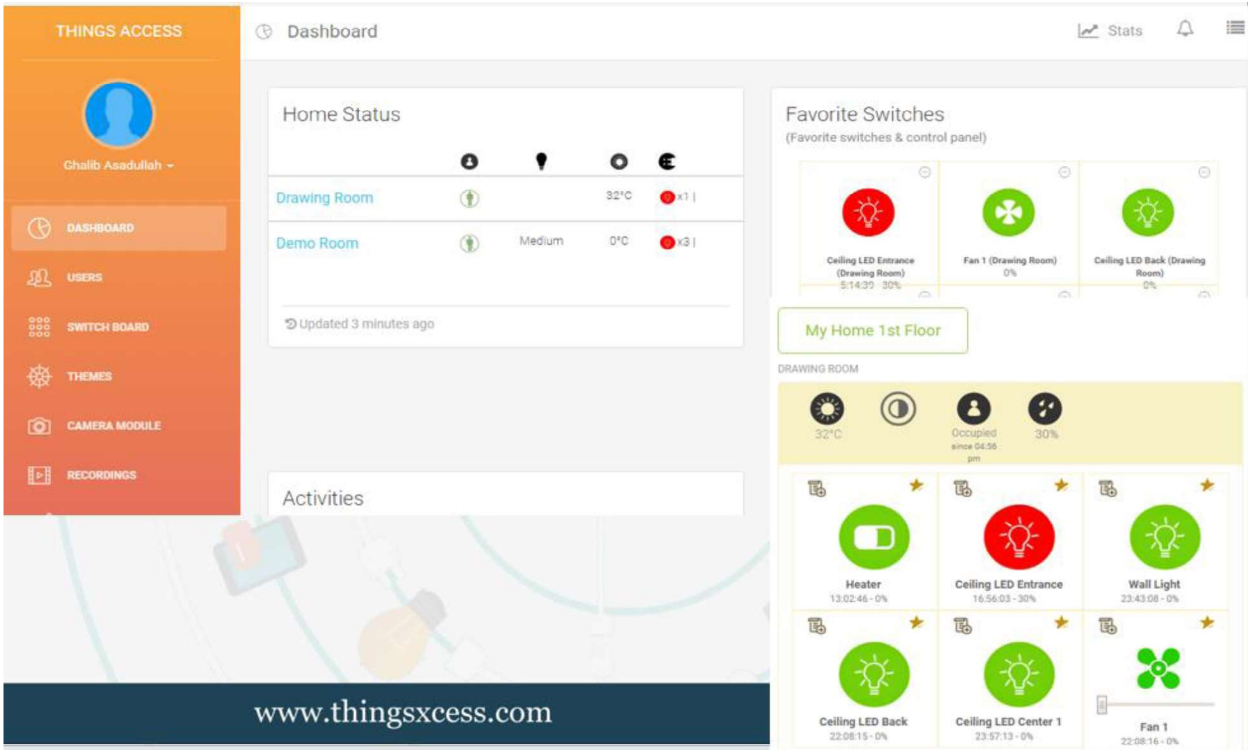

Fig. 1.1 User Interface of a Smart Home

> # Teacher Notes:
>
> - Give real-life live examples of IoT systems to students.

Figure 1.2 shows an example of an IoT system.

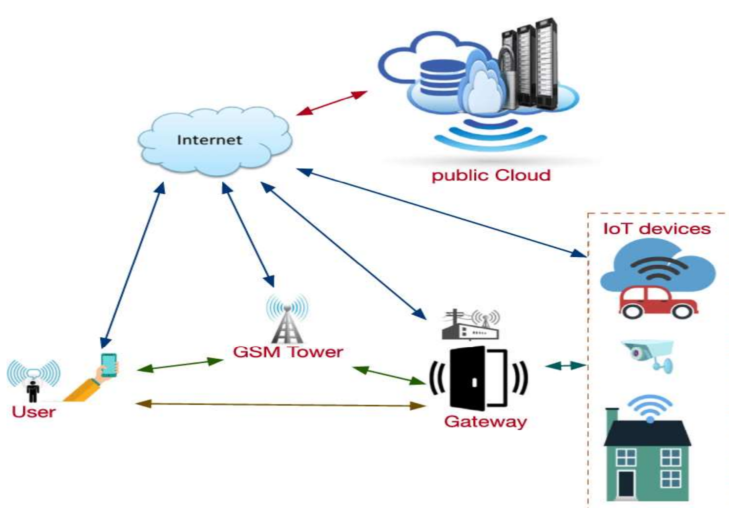

Fig. 1.2 Example of an IoT System

1.1.2 Importance of IoT

Consider an IoT use case: you went to the market and forgot to switch off your AC, fan, or light and felt helpless that you cannot return home to switch it off? This is where IoT comes into the picture. It can remind you of the essential tasks that you often forget to do. Also, using the mobile application, you can access your home appliances integrated with IoT from anywhere around the world.

Another very exciting IoT device is the NFC (near field communication) smart ring.These rings serve as a multi-purpose device. They are rings consisting of the connectivity of the network, NFC chips, and sensors that help you exchange data. By using NFC rings, you can pay your bills, access your car's door lock by just swiping it, and also get your mobile notifications. ANFC ring is shown in Figure 1.3.

Fig. 1.3 Example of NFC Ring

All this is possible due to the connection of the IoT devices over the internet servers that helps in sharing and exchanging the data. This technology reduces human efforts and saves a lot of time.

1.2 Scope of IoT

Internet of Things has emerged as a leading technology around the world. It has gained a lot of popularity in lesser time. Also, the advancements in Artificial Intelligence and Machine Learning have made the automation of IoT devices easy. Basically, AI and ML programs are combined with IoT devices to make them intelligent. Due to this, IoT has also expanded its area of application in various sectors. IoT applications are found in smart homes, smart buildings, smart cities, smart agriculture and smart transportation etc. Figure 1.4 enlists scope of IoT in different domains.

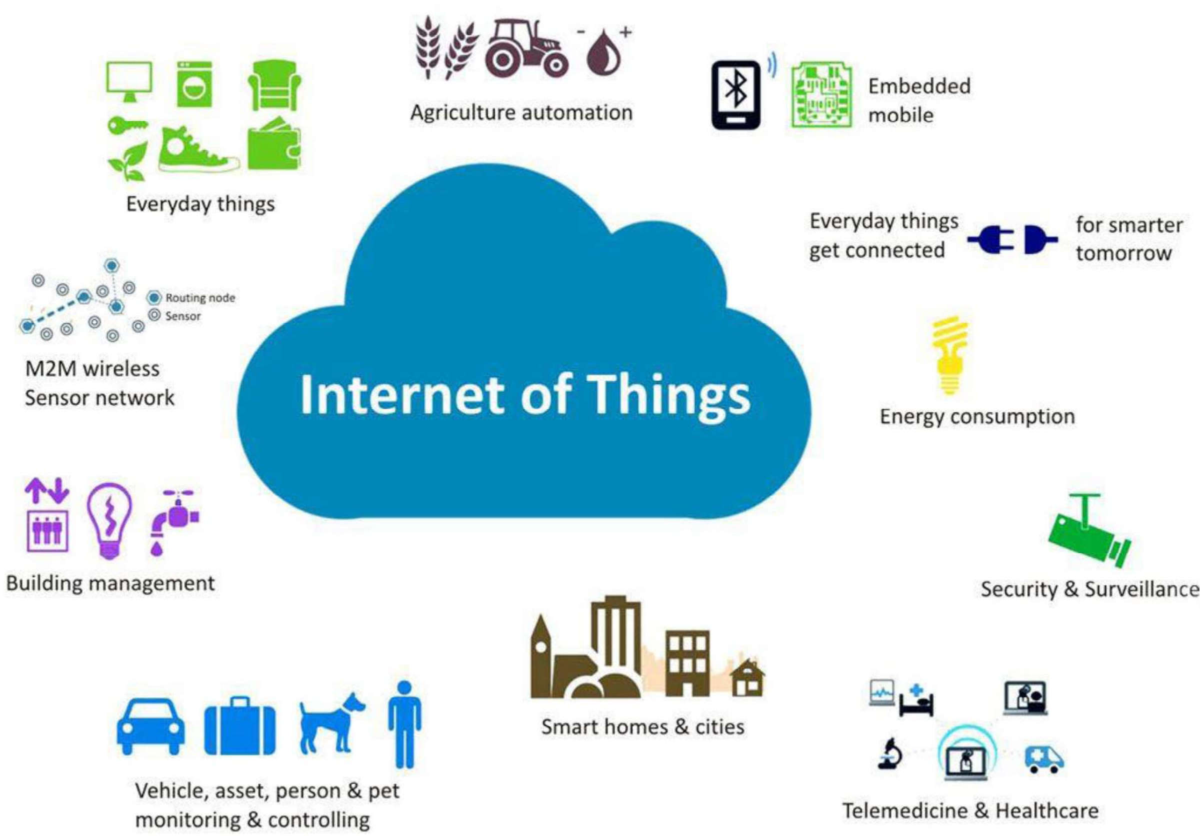

Fig. 1.4 Scope of IoT

> **Do you know?**
> There are expected to be more than 30.9B IoT devices worldwide by 2025. According to the latest research, the number of IoT-connected devices globally reached 11.7 billion in 2020.

1.3 Verticals of IoT

The following are some IoT vertical markets that have most noticeably adopted IoT solutions to date.

1.3.1 Smart Buildings

Smart building management involves collecting data from smart devices and sensors to remotely monitor a property's energy, security, landscaping, HVAC (heating, ventilation and control), lighting and more. Actions can be automated according to events and efficiency can be optimized, saving time, resources and cost.Figure 1.5 shows components of a smart building.

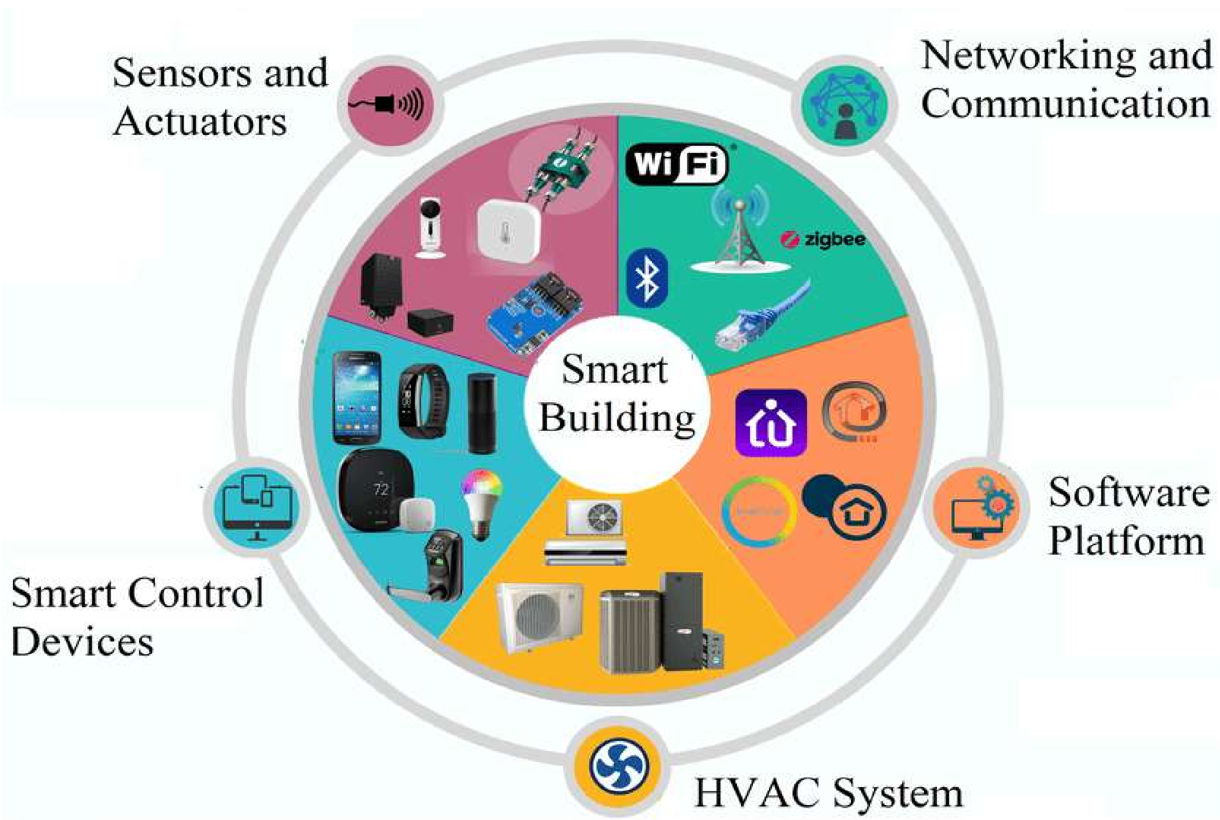

Fig. 1.5 Components of a Smart Building

1.3.2 Industrial IoT

Industrial IoT (IIoT) uses system integration and sensors to gather data within a process for analysing and optimizing different manufacturing and other processes. As a result, human error and operation cost is reduced. For example, as operation costs fluctuate in oil and gas sector, remote monitoring and insightful decision making can keep an enterprise in the oil andgas industry successful. Regulatory compliance can be accurately monitored and overall costs decrease.

Agriculture is one of the major industries constituting a large sector of economy. It involving food farming, livestock farming and cotton farming. With the ever-increasing population of the earth, it is important for the farming industry to operate as efficiently and effectively as possible. IoT can enable local and commercial farming to be more environmentally friendly, cost effective and production efficient. Figure 1.6 shows components of Industrial IoT.

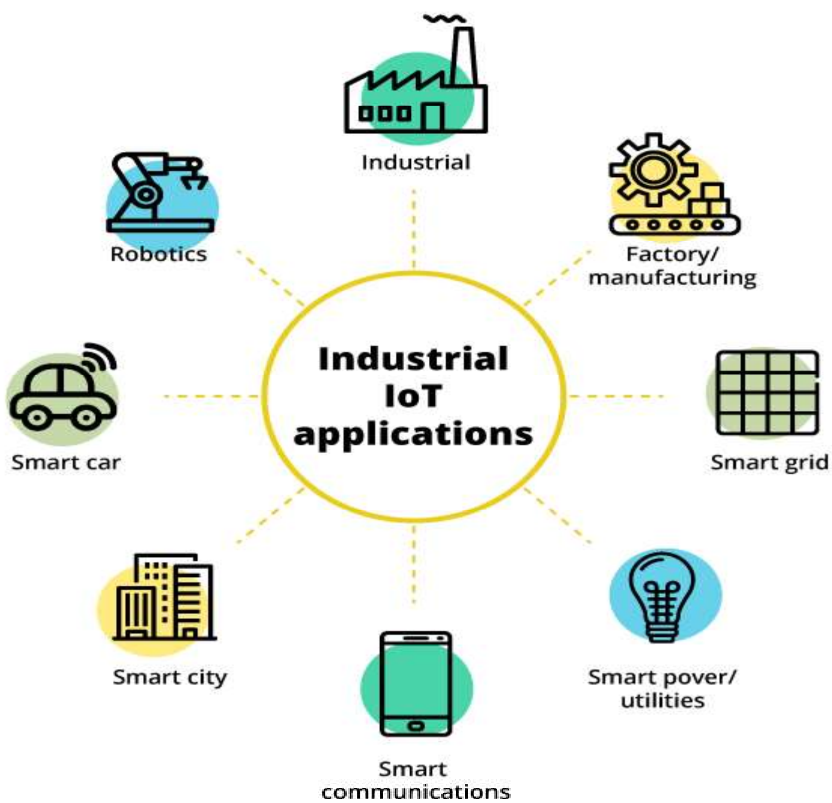

Fig. 1.6 Components of Industrial IoT

1.3.3 Smart Transportation and Logistics

IoT can also be employed in smart transportation and logistics management. In an IoT enabled smart transportation environment, fleet management, asset tracking and predictive maintenance work together as an end-to-end solution. Figure 1.7 shows components of a smart transportation system.

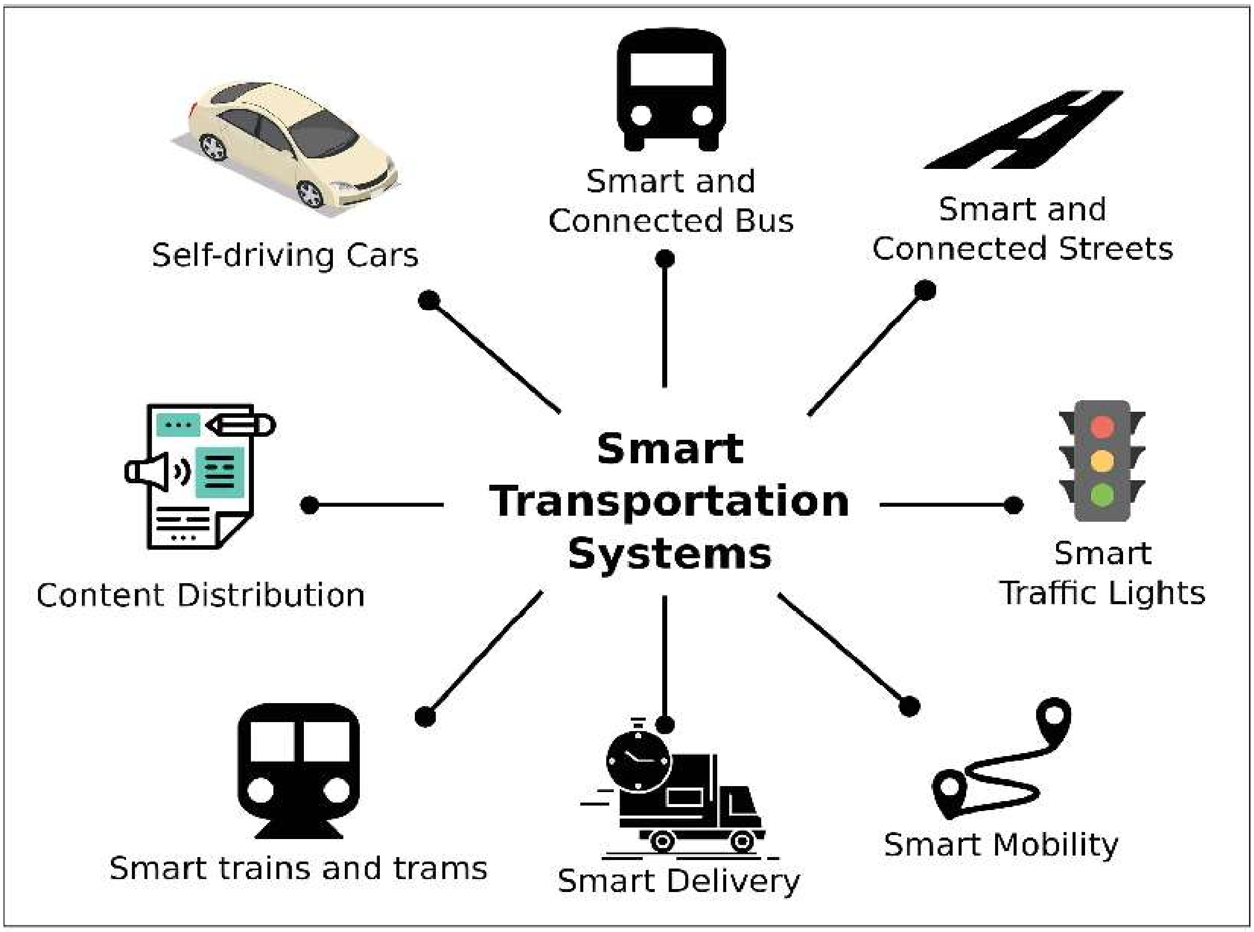

Fig. 1.7 Components of Smart Transportation

1.3.4 Smart Homes

Consumers connecting smart devices within homes can enhance the home experience, increase home security and conserve energy. Homeowners can now monitor their properties remotely and in real-time. Figure 1.8 shows components of a smart home.

Do you know?
The Google Smart Home platform lets users control your commercially available connected devices through the Google Home app and Google Assistant, which is available on more than 1 billion devices, like smart speakers, phones, cars, TVs, headphones, watches, and more.

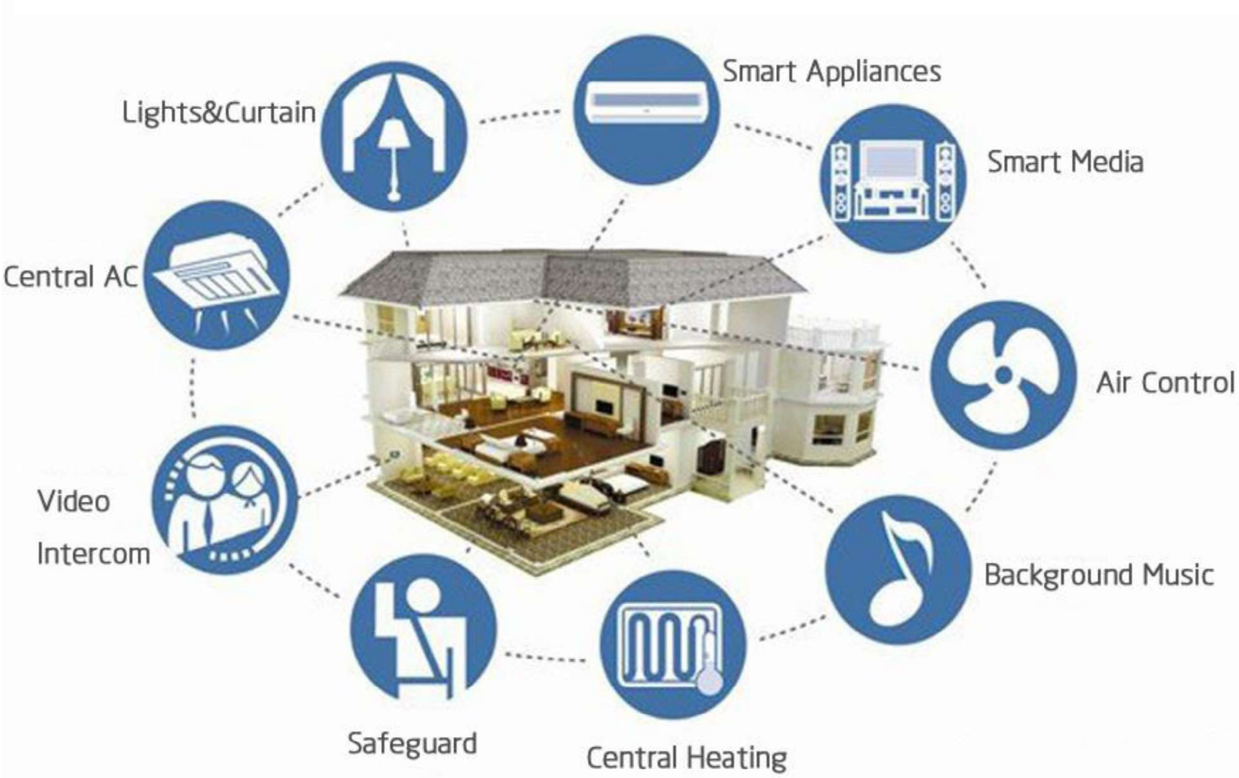

Fig. 1.8 Components of a Smart Home

1.3.5 Smart Cities

A smart city can include anything from smart parking to mass transit. A smart city addresses traffic, public safety, energy management and more for its government and citizens. More and more cities across the world are adopting these solutions at a steady rate. Figure 1.9 shows components of a smart city.

Interesting Information
- Singapore is the smartest city in the world, according to the IMD's inaugural Smart City Index.

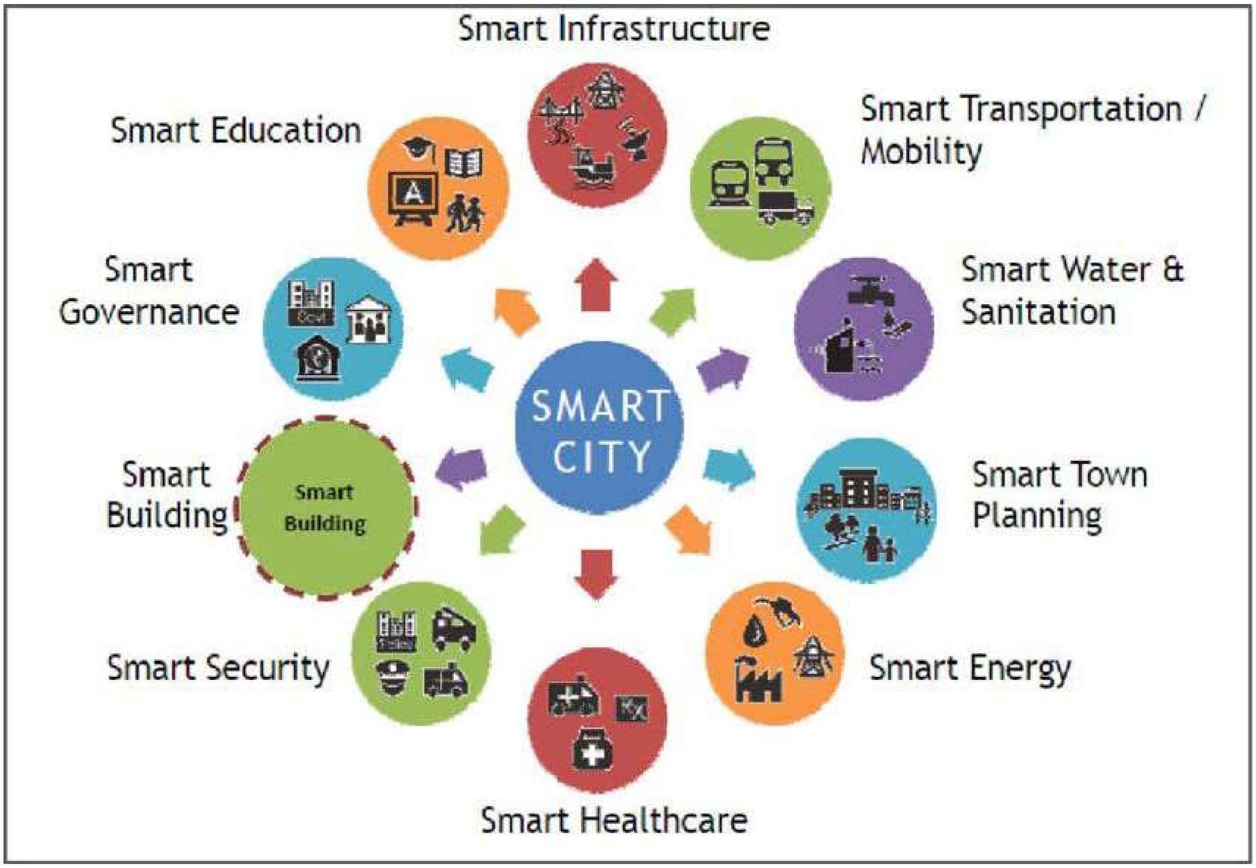

Fig. 1.9 Components of a Smart City

1.3.5 Smart Healthcare

Healthcare is perhaps one of the fastest adopters of smart, connected technology. IoT enables critical business and patient monitoring decisions to be made remotely and in real-time. Figure 1.10 shows components of a smart healthcare system.

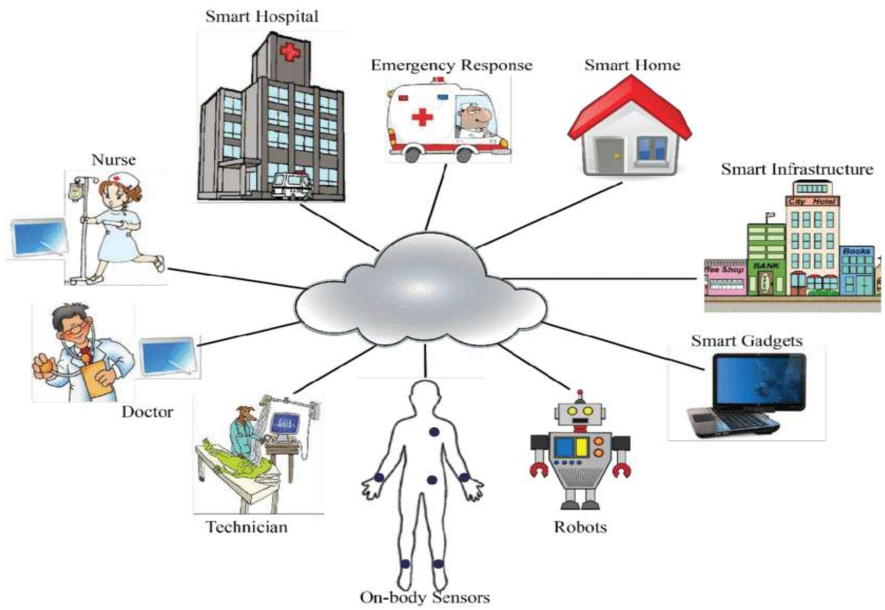

Fig. 1.10 Components of Smart Healthcare

These are just a few of the many vertical markets adopting Internet of Things solutions in the world today. Many of these solutions have no doubt had an effect on you, whether noticed or not. In fact, the seamless experience of these solutions is actually one of the most important aspects of IoT implementation and adoption by consumers. The acceleration of the internet of things will only be more prevalent moving forward.

Key Points

- The Internet of Things (IoT) refers to the billions of physical devices around the world that are connected to the Internet.
- An IoT system consists of Sensors, actuators, connectivity medium, IoT cloud and a user interface.
- IoT applications are found in smart homes, smart buildings, smart cities, smart agriculture and smart transportation etc.
- Smart building management involves collecting data from smart devices and sensors to remotely monitor a property's energy, security, landscaping, HVAC, lighting and more.
- Industrial IoT (IIoT) uses system integration and sensors to gather data within a process for analysing and optimizing different manufacturing and other processes.
- In an IoT enabled smart transportation environment, fleet management, asset tracking and predictive maintenance work together as an end-to-end solution.
- Consumers connecting smart devices within homes can enhance the home experience, increase home security and conserve energy.
- A smart city addresses traffic, public safety, energy management and more for its government and citizens.
- In healthcare, IoT enables critical business and patient monitoring decisions to be made remotely and in real-time.

Exercise

Select the most appropriate option

1. IoT stands for

   a. Internet of Devices b. Internet of triangles c. Internet of Trees d. Internet of Things

2. There will be around ______ billion IoT devices by 2025.

   a. 40 b. 20 c. 25 d. 30

3. Fleet managemnet lies in which IoT vertical?

   a. Smart Cities b. Smart Agriculture c. Industrial IoT d. Smart transportation

4. HVAC is a main component of which IoT vertical

   Smart Cities b. Smart Agriculture c. Industrial IoT d. Smart transportation

5. Which is the smartest city in the world

   a. Singaporeb. Helsinki c. Tokyo d. None of these

Write short answer of the following

1. Define IoT.
2. What is the importance of IoT?
3. What is the scope of IoT?
4. Enlist some applications of IoT.

Answer the following question in detai.

1. Explain different components of an IoT system.
2. Describe IoT verticals and their components.